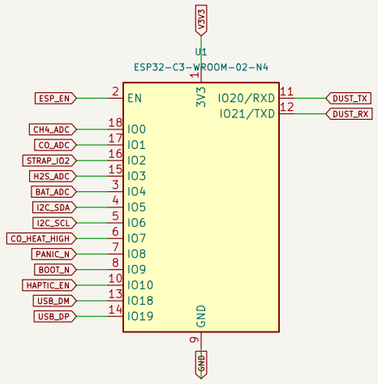
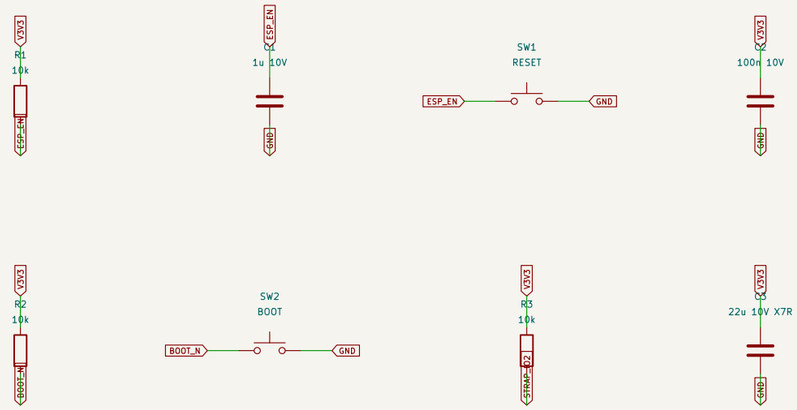
ESP32-C3-WROOM-02-N4 | all physical pins assigned

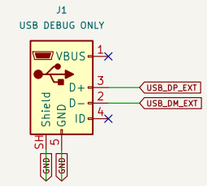


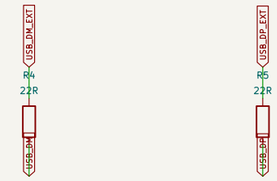
ADC1 only: GPIO0/1/3/4. GPIO5 is ADC2 and used as I2C SDA.
GPIO2 is reserved with a pull-up. GPIO8 panic held HIGH for download.
GPIO18/19 are native USB. UART0 GPIO20/21 serves the dust sensor.



EN: 10k / 1uF RC. BOOT: hold SW2, pulse RESET, release SW2.
Release the PANIC button while entering download mode.
Keep module antenna clear of copper, battery, housing metal and wiring.

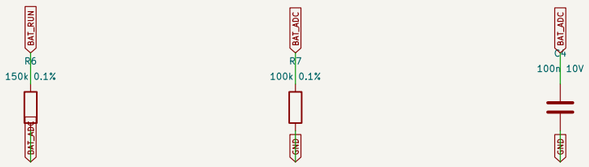
Native USB debug | battery powered in RUN





J1 VBUS is intentionally NC; it cannot power or charge the node.
Route D+/D- as a 90-ohm differential pair; 22R at the MCU.
Add validated USB ESD parts in PCB design; no EMC claim is made.
Do not use UART0 ROM chatter as valid dust-sensor commands.

Battery voltage monitor | 0 to 4.25 V pack



VADC = 0.4 x VBAT: 4.25 V becomes 1.70 V.
Use calibrated ADC1 attenuation 12 dB, nominal range up to 2.5 V.
Firmware low-battery stop target: 3.2 V loaded, with hysteresis.
Protected pack cutoff is the final battery safeguard, not firmware.

Other sheets contain the power, sensing and alert circuitry.
Same global net names are electrically connected across sheets.

Protected battery, chargi	Gas sensor conditioning a	Dust, motion, display, pa
File: Power.kicad_sch	File: Gas_Sensors.kicad_sch	File: User_And_Environment.kicad_sch

NOT INTRINSICALLY SAFE — NOT FOR EXPLOSIVE ATMOSPHERES

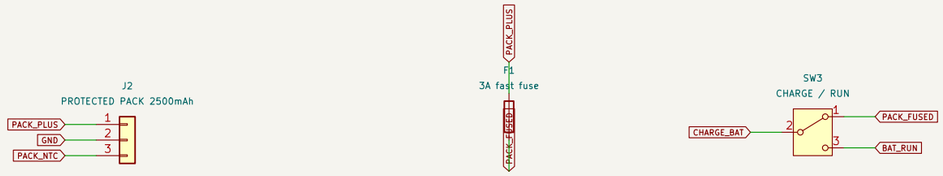
PRAN RAKSHAK | VK26-IITH-SM-0275

Sheet: /
File: PRAN_RAKSHAK.kicad_sch

Title: 01 | ESP32-C3 controller and programming

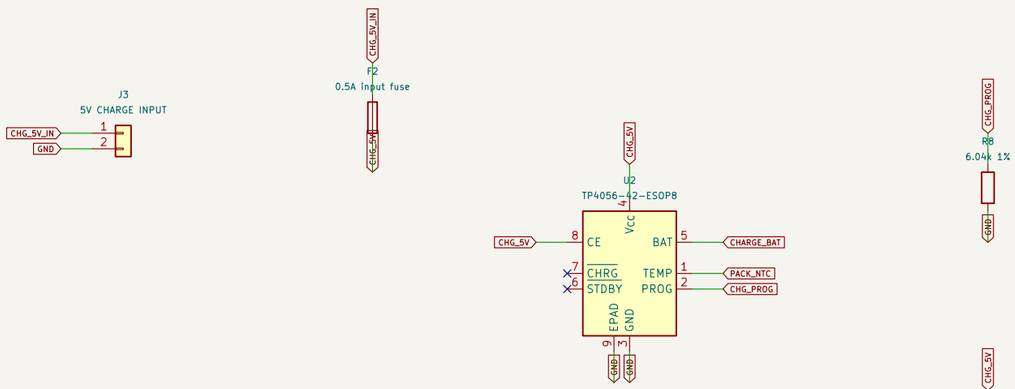
Size: A3	Date: 2026-10-25	**Rev: A — BENCH PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 1/4

1S protected LiPo | charge OR run, never both



SW3 common pin 1 -> fused pack; throw 2 = CHARGE; throw 3 = RUN.
Break-before-make, rated >=3 A DC. All node loads use BAT_RUN.
Protected pack is mandatory; TP4056 itself does NOT protect discharge.
Charge only in a non-hazardous area with the node disabled.

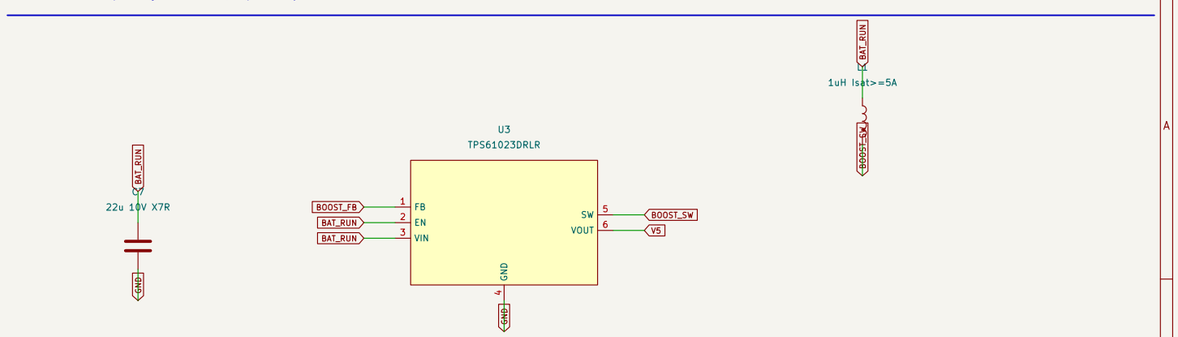
TP4056 charger | approx 180–200mA | cell NTC



6.04k sets approx 182..199mA by IC revision; verify actual current.
Worst charger heat approx (5.25–3.0)*0.2 = 0.45 W.
NTC open/hot/cold must inhibit; a short can disable TEMP protection.
Cell-mounted NTC and charge fault tests are mandatory before use.



5 V boost | adjustable, qualify heater tolerance



TPS61023 VFB typ 0.595V, VOUT = VFB x (1 + Rtop/100k).
Trim 5.000V before fitting gas sensors. Require 4.90 to 5.10V
at sensor pins over load, battery and temperature; not guaranteed
by nominal resistors. Budget 1.2A peak at 5V; validate thermals.

3.3 V linear regulator | heat-spreader required



At 5V input: Pd = 1.7 x Iout; 0.25A -> 0.425W, 0.5A -> 0.85W.
SOT-223 copper area / temperature rise must be verified on PCB.

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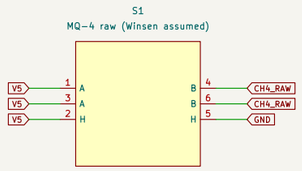
PRAN RAKSHAK | VK26-IITH-SM-0275

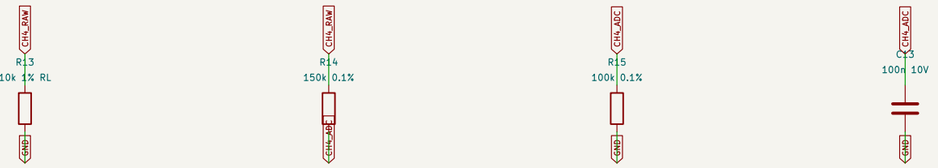
Sheet: /Protected battery, chargi/
File: Power.kicad_sch

Title: 02 | Protected battery, charging and regulated rails

Size: A3	Date: 2026-10-25	**Rev: A – BENCH PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 2/4

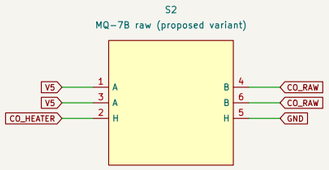
MQ-4 methane | raw sensor, continuous heater

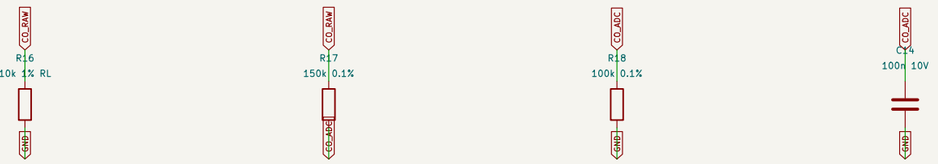




Heater: 5.0V +/-0.1V. <=1W. Do not arbitrarily duty-cycle.
A/B electrodes are not interchangeable with the heater terminals.
48h initial preheat per manufacturer; calibrate with traceable gas.
VADC=0.4 x VRAW, max 2.10V at 5.25V; RC pole approx 26.5Hz.

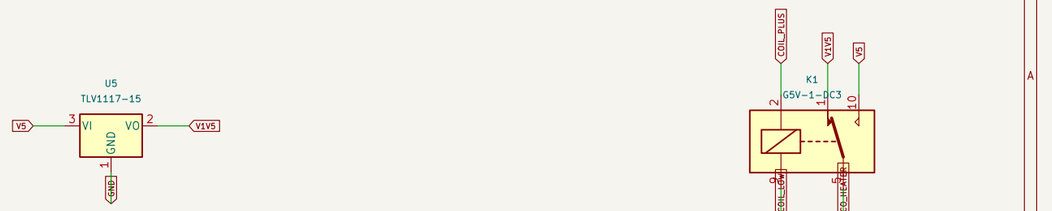
MQ-7 CO | sensing circuit always on 5V

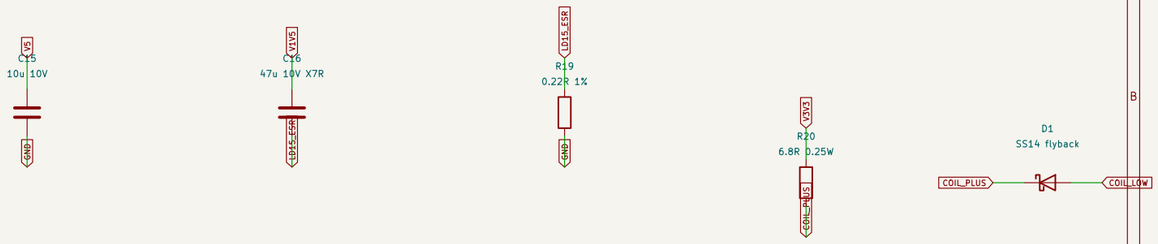




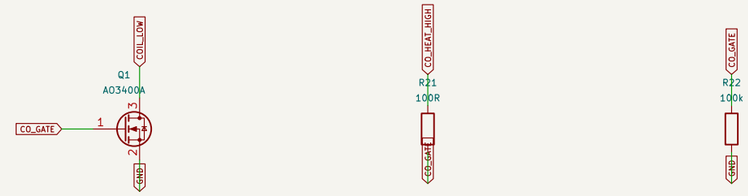
Sample calibrated CO only in the specified low-temperature phase.
A 1Hz ADC read is NOT a 1Hz independent calibrated CO result.

Heater selector | electromechanical BENCH solution

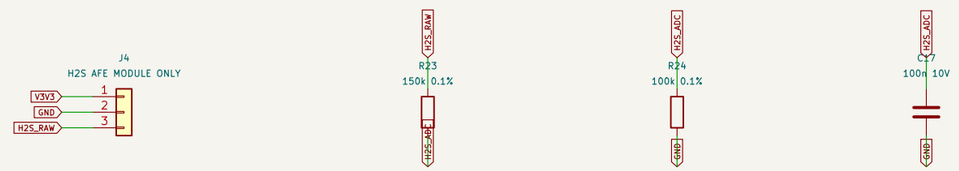




GPIO7 HIGH: 5V heater for 60s; LOW: 1.5V heater for 90s.
Default/reset = low heat. Firmware must complete cleaning on startup.
Relay isolates the two rails without backfeeding the 1.5V regulator.
Relay contact arcing and hot MQ heaters prohibit hazardous-area use.



H2S conditioned module interface | model unresolved



NO raw cell connection. Exact model, bias, sensitivity, zero-current,
TIA gain, polarity, range and calibration remain release blockers.
3.3V output -> 1.32V ADC; clamp/ESD design depends on final module.

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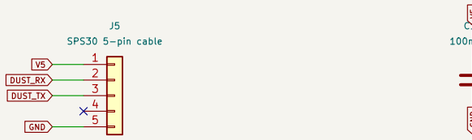
PRAN RAKSHAK | VK26-IITH-SM-0275

Sheet: /Gas sensor conditioning a/
File: Gas_Sensors.kicad_sch

Title: 03 | Gas sensor conditioning and MQ-7 heater cycle

Size: A3	Date: 2026-10-25	**Rev: A — BENCH PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 3/4

Dust sensor | SPS30 proposed; UART0 3.3V logic





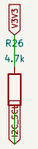


SPS30 supply 4.5..5.5V; 65mA max measuring, 80mA fan startup.
TX high 2.9..3.37V supports ESP32-C3; RX accepts 3.3V.
Disable UART0 application console; account for ROM boot chatter.
Its airflow opening and fan challenge the PDF size/IP67 targets.

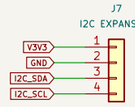
3.3V I2C OLED and expansion



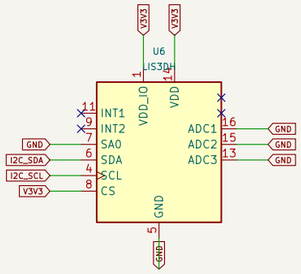




One pair of pullups only. 100kHz default; qualify rise times.
Optional field runoff/water sensors need separate electrical design.
They are not defined by the PDF and are not direct-wired here.



Man-down sensor | LIS3DH, I2C address 0x18





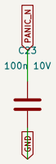




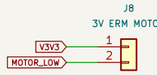
Poll at 25..50Hz, compute motion/orientation and time no-motion.
A 30s inactivity condition needs validation against real worker activity.
This schematic supplies sensing hardware, not fall-detection firmware.

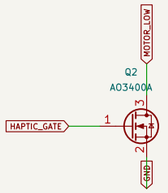
Panic and haptic | local warning

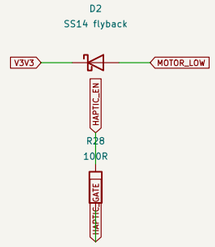


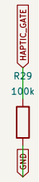


GPIO8 is a strap. Panic must be released for USB download:
SPI boot accepts either GPIO8 level when GPIO9 stays high.
Debounce in firmware and transmit manual emergency events.









Motor power from 3.3V regulator. Flyback cathode to V3V3.
GPIO10 HIGH turns motor on; gate pulldown keeps it off during reset.

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Sheet: /Dust, motion, display, pa/
File: User_And_Environment.kicad_sch

Title: 04 | Dust, motion, display, panic and haptic alert

Size: A3	Date: 2026-10-25	**Rev: A – BENCH PROTOTYPE**
KiCad E.D.A. 10.0.7		Id: 4/4